



10149 - The physical drivers of burstiness - a deep dive on LULLing galaxies below the main sequence at $z > 4$

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Plan_0	NIRSpec MultiObject Spectroscopy	(1) LULLS_catalog

ABSTRACT

The physical processes regulating star formation in low-mass galaxies that dominate the early Universe ($z > 4$) are to date poorly understood. These systems are theorized to undergo stochastic cycles of bursting and “dormant” phases, but the specific feedback mechanisms responsible are difficult to constrain observationally. Current theoretical frameworks typically assume that stellar feedback - such as stellar winds or supernovae - alone regulates this burstiness. However, this is in direct tension with recent observations confirming the widespread presence of low-luminosity accreting massive black holes (AGN) at these epochs, whose influence is largely excluded from existing models. To resolve this critical gap, we must directly probe the physical state of the gas in galaxies where star formation has been recently suppressed. We therefore propose the first grating spectroscopy survey of 15 low-mass, low-SFR (“LULLing”) galaxies at $z=4-8$ that lie significantly below the star-forming main sequence (SFMS) - systems where feedback is expected to have recently driven the transition toward suppressed activity. Building on deep PRISM observations that securely identified these galaxies as weak line emitters, we will use JWST/NIRSpec G235M and G395M for a total of 18+60 h on source (97 h total). These observations will enable precise measurements of gas kinematics (line widths), photoionization diagnostics, and metallicities. By targeting galaxies at different stages below the SFMS - precisely where feedback is expected to have suppressed star formation - we will, for the first time, directly test whether stellar or AGN-driven processes regulate burstiness in the early Universe.

OBSERVING DESCRIPTION

We propose ultra-deep JWST/NIRSpec grating spectroscopy of faint, low-mass galaxies lying significantly below the star-forming main sequence to investigate the physical mechanisms driving feedback, black-hole activity, outflows, and shocks in the early Universe. The observations will target narrow and broadened components of key rest-frame optical emission lines — H, H, [O III], [O II], and fainter diagnostics — to constrain ionization conditions, kinematics, and excitation sources. We request one NIRSpec/MSA pointing using the G235M and G395M gratings, totaling 18 h + 60 h on-source integration time. Simultaneous ultra-deep NIRCам parallel imaging in complementary wide- and medium-band filters will provide rest-frame UV–optical continuum constraints and improved photometric coverage for SED modeling.

Proposal 10149 - Targets - The physical drivers of burstiness - a deep dive on LULLing galaxies below the main sequence at $z > 4$

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	LULLS_catalog	RA: 03 32 30.5509 (53.1272954d) Dec: -27 49 21.29 (-27.82258d) Equinox: J2000		
Comments: Description=[]					

Proposal 10149 - Observation 1 - The physical drivers of burstiness - a deep dive on LULLing galaxies below the main sequence at z ...

Observation	Proposal 10149, Observation 1: Plan_0											Fri Mar 13 19:00:37 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	LULLS_catalog	RA: 03 32 30.5509 (53.1272954d) Dec: -27 49 21.29 (-27.82258d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	4		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	5		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	6		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCам Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: Targets (19 sources)											
	Filler Candidate List: Fillers (242 sources)											
	Spectral Overlap Map: jwst-nirspec-mr											
Spectral Overlap Threshold: 1.5												

Dithers	Reference Stars	
Dithers	#	Dither Type
	1	NONE

Proposal 10149 - Observation 1 - The physical drivers of burstiness - a deep dive on LULLing galaxies below the main sequence at z ...

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	53.180404208333 33 Degrees - 27.773691666666 65 Degrees	99.545377052315 64			3	15	21008.002
	2	2 (G395M/F290LP)	c1	3 Shutter Slitlet	53.180404208333 33 Degrees - 27.773691666666 65 Degrees	99.545377052315 64			3	15	21008.002
	3	3 (G395M/F290LP)	c1	3 Shutter Slitlet	53.180404208333 33 Degrees - 27.773691666666 65 Degrees	99.545377052315 64			3	18	25209.602
	4	4 (G395M/F290LP)	c1	3 Shutter Slitlet	53.180404208333 33 Degrees - 27.773691666666 65 Degrees	99.545377052315 64			3	18	25209.602
	5	1 (G235M/F170LP)	c2	3 Shutter Slitlet	53.180474 Degrees - 27.773325277777 758 Degrees	99.545345398943 72			3	15	21008.002
	6	2 (G395M/F290LP)	c2	3 Shutter Slitlet	53.180474 Degrees - 27.773325277777 758 Degrees	99.545345398943 72			3	15	21008.002
	7	3 (G395M/F290LP)	c2	3 Shutter Slitlet	53.180474 Degrees - 27.773325277777 758 Degrees	99.545345398943 72			3	18	25209.602
	8	4 (G395M/F290LP)	c2	3 Shutter Slitlet	53.180474 Degrees - 27.773325277777 758 Degrees	99.545345398943 72			3	18	25209.602
	9	1 (G235M/F170LP)	c3	3 Shutter Slitlet	53.180334333333 33 Degrees - 27.774058055555 543 Degrees	99.545408746045 48			3	15	21008.002
	10	2 (G395M/F290LP)	c3	3 Shutter Slitlet	53.180334333333 33 Degrees - 27.774058055555 543 Degrees	99.545408746045 48			3	15	21008.002
	11	3 (G395M/F290LP)	c3	3 Shutter Slitlet	53.180334333333 33 Degrees - 27.774058055555 543 Degrees	99.545408746045 48			3	18	25209.602
	12	4 (G395M/F290LP)	c3	3 Shutter Slitlet	53.180334333333 33 Degrees - 27.774058055555 543 Degrees	99.545408746045 48			3	18	25209.602

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	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
Spectral Elements	1	F210M	F277W	DEEP8	7	5	15	3	20743.44	
	2	F210M	F356W	DEEP8	7	5	15	3	20743.44	
	3	F150W	F277W	DEEP8	7	6	18	3	24898.57	
	4	F150W	F356W	DEEP8	7	6	18	3	24898.57	
	5	F210M	F360M	DEEP8	7	5	15	3	20743.44	
	6	F210M	F410M	DEEP8	7	5	15	3	20743.44	
	7	F150W	F360M	DEEP8	7	6	18	3	24898.57	
	8	F150W	F410M	DEEP8	7	6	18	3	24898.57	
	9	F210M	F444W	DEEP8	7	5	15	3	20743.44	
	10	F150W	F444W	DEEP8	7	5	15	3	20743.44	
	11	F150W	F430M	DEEP8	7	6	18	3	24898.57	
	12	F150W	F460M	DEEP8	7	6	18	3	24898.57	
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 98.57 to 100.57 Degrees (V3 319.9954303 to 321.9954303) Visits Same PA No Parallel Attachments MSA Planned Aperture PA 99.5700 to 99.5700 Degrees (V3 320.9954303 to 320.9954303)									